

MID-SEMESTER EXAMINATION, November-2023

Introduction to Theory of Computation (CSE3731)

Programme: B.Tech(CSE/CSIT)

Semester: 5th

Full Marks: 30

Time: 2 Hours

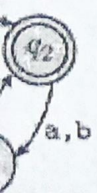
Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
CO1: Enhance/develop ability to understand and conduct mathematical proofs for computation and algorithms.	L1,L3	1	2+2+2
CO2: Design and analyze finite automata and regular expression for describing regular languages.	L2,L6	2	2+2+2
	L2,L6	3	2+2+2
	L3,L6	4	2+2+2
CO3: Design and analyze pushdown automata, and context-free grammars.	L2,L6	5	2+2+2

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries equal mark.

1.	(a)	Define the term Automaton. Describe its characteristics with the help of a suitable model.	2
	(b)	Definition: A graph G is said to be k -regular if every node in the graph has degree k . Applying proof by construction, prove that "For each even number n greater than 2, there exists a 3-regular graph with n nodes".	2
	(c)	Definition: Let g and f be functions from the set of natural numbers to itself. The function $f = O(g(n))$ (read as big-oh of g of n), if there is a constant $c > 0$ and a natural number n_0 such that $f(n) \leq cg(n)$ for all $n \geq n_0$. In computational complexity theory, let two algorithms run in a machine at time 2^{2n} and 2^n . Applying proof by contradiction, prove that $2^{2n} \neq O(2^n)$.	2
2.	(a)	Differentiate between Deterministic Finite Automata (DFA) and Non-deterministic Finite Automata (NFA).	2

	(b)	Design a deterministic finite Automaton (DFA) for the language accepting all the strings that ends with 3 a's or 3 b's over input alphabets $\Sigma = \{a, b\}$.	2
	(c)	Construct the state transition diagram of the deterministic finite Automaton (DFA) for the language accepting all the strings of length 4 over $\Sigma = \{0, 1\}$ such that every string contain atleast two 0's .	2
3.	(a)	Construct an NFA for the language accepting all strings over $\Sigma = \{0, 1\}$ such that each string contains a 1 (one) in the 3 rd position from the end.	2
	(b)	Construct an equivalent DFA for the NFA obtained from Question No. 3(a).	2
	(c)	Identify the languages recognized by the given finite state machines M1 and M2.	2
		M1 M2	
4.	(a)	Derive the regular expression for the given finite state automaton.	2
	(b)	Using Pumping lemma prove the following language is not regular. $L = \{0^n 1^n \mid n \geq 0\}$	2
	(c)	Construct the finite automaton equivalent to the regular expression $R = 10 + (0 + 11) 0^* 1$ [Here the operators * (star) represents the Kleene-closure or Kleene-star, + (plus) represents the Union and a string in the form "ab" represents concatenation of symbols a and b]	2
5.	(a)	Let the context-free grammar G be given by the productions $\{S \rightarrow AB, A \rightarrow a, B \rightarrow b, B \rightarrow C, E \rightarrow c\}$. Find the context-free grammar G_1 such that every variable in G_1 derives some terminal string.	2

the language s over input	2
deterministic ing all the ring contain	2
strings over e) in the 3 rd	2
ained from	2
finite state 	2
finite state 	2
uage is not	2
ular	2
of symbols	
uctions context-free erives some	2

(b) ✓	Construct a grammar in Chomsky normal form(CNF) from the equivalent CFG, G, with production rules $S \rightarrow aAbB$, $A \rightarrow aA a$, $B \rightarrow bB b$.	2
(c) ✓	If $G = (\{S, A\}, \{a, b, \epsilon\}, P, S)$ where P consists of $S \rightarrow aAS aSS \epsilon$, $A \rightarrow SbA ba$. Show that the grammar G is ambiguous for the string aabaa .	2
	End of Questions	